

PAMM: Pathway-Aware Masked Representation Learning for Interpretable Multi-Cancer Prediction

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Abstract—The multi-cancer prediction with high-dimensional transcriptomics presents serious issues because of the small n large p problem, large noise rates, and low biological explainability of the traditional deep learning models. Current self-supervised and supervised methods typically use stochastic masking or black-box representations, and these do not describe functional biological interactions in pathways of genes. In response to these shortcomings, this paper recommends an interpretable multi-cancer prediction framework named PAMM that is premised on Pathway-Aware Masked Representation Learning. The model uses biologically motivated priors based on the KEGG 2021 human pathway database in which full functional sets of genes are masked during pretraining and the networks are free to form missing biological units and discover relations between pathways. The pre-processing pipeline is done rigorously to alleviate noise and dimensionality in samples of Breast, Lung, Glioblastoma, and Hepatocellular Carcinoma. Optuna is used to optimize the latent representations and the representations are tested by repeating the process several times and ensuring the statistical robustness. The interpretation of the models is done through single-sample Gene Set Enrichment Analysis (ssGSEA), which provides unique and clinically meaningful patterns of pathways activity in each of the cancer types. Besides closed-set classification, PAMM is open-set recognizing that it analyzes both softmax confidence and latent-space distances, which can be used to reject unknown or non-cancerous samples. Experimental findings indicate that PAMM is able to detect biologically significant cancer-specific signatures in addition to offering a secure and dependable diagnostic framework to enable precise oncology.

Index Terms—Pathway-Aware Masking, Self-Supervised Learning, Multi-Cancer Prediction, Interpretable AI, KEGG Pathways, ssGSEA, Open-Set Recognition

I. INTRODUCTION

Modern oncology has changed the traditional diagnosis of morphology to a high-resolution molecular profiling with the introduction of high-throughput sequencing (HTS), the next-generation sequencing (NGS). Transcriptomics measures the expression level of the genome in RNA and gives a dynamic picture of cell activity, which is a response to oncogenic transformation, environmental stress, and treatment. Transcriptomic profiles can be considered the future of precision oncology as they can record regulatory and pathway-scale perturbations that promote tumorigenesis, metastasis, and resistance to therapy, unlike the comparatively stable genome.

Transcriptomic datasets have extremely high dimensions, which is despite their potential; they have more than 50,000 gene features, and they are measured in a few samples. Such an imbalance produces the so-called small n large p problem, which limits the ability of traditional statistical and machine learning piping. Although initial applications of systems biology concentrated on individual differentially expressed genes, it has been determined that genes function in concerted functional modules. This means that prediction of multi-cancer can only be done reliably by models that can take advantage of the organization of pathways and the ability to model coordinated biological behavior.

The recent deep learning methods have recorded high performance in cancer classification, but due to the black-box nature, it is a great challenge in their application in clinical practice. Neural networks like multilayer perceptrons and convolutional networks typically offer an accurate prediction without biological explanations, and hence, cannot be trusted or acted upon clinically. Furthermore, a majority of transcriptomic classifiers are based on closed-set environment, which requires the classification of all samples into pre-defined cancer classes. In practice, in clinical practice, samples can be of rare, mixed or new forms of cancer whose overconfident prediction can be clinically dangerous.

In order to eliminate these shortcomings, this paper suggests PAMM (Pathway-Aware Masked Representation Learning), a framework of interpretable and robust multi-cancer predictions. PAMM combines biological priors based on the KEGG 2021 Human pathway database in a self-supervised masked learning model. Rather than random masking, entire functional gene modules are masked during pretraining and they promote the model to learn inter-pathway dependencies. Multi-stage preprocessing pipeline minimizes dimensionality and overfitting, and open-set recognition is supported by analysis of latent space and rejection based on confidence. These contributions, combined, will bring high-dimensional self-supervised learning to high-dimensional biologically based and clinically trustworthy cancer diagnostics.

II. RELATED WORK

The last several years were marked by the active research on the problem of machine learning and deep learning applied

to genomic, transcriptomic, and clinical data to predict and classify cancer. Available literature varies in terms of learning paradigm, biological prior integration, level of supervision and interpretability. In this part, the most appropriate literature has been reviewed by classifying the previous works as either traditional machine learning methods, methods based on deep learning, methods based on unsupervised and clustering, information-based methods, hybrid methods based on learning and new developments in the fields of generative and self-supervised learning. The shortcomings of these methods are the driving force behind the suggested Pathway-Aware Masked Representation Learning (PAMM) system.

A. Conventional Prediction of cancer using machine learning

Initial research used classical machine learning (ML) methods to predict cancer in terms of clinical, genomic, and image data. As it was shown by Kourou et al. (2015) [1], artificial neural networks (ANNs), support vector machines (SVMs), decision trees, and Bayesian networks are all the models that enhance early cancer prognosis and customized treatment. Nonetheless, clinical adoption was prohibited by small samples, heterogeneous data and overfitting. It was demonstrated that genetic data and lifestyle factors contribute significantly to the estimation of cancer risk in 20 cancer types, as reported by Kim et al. (2018) [2] who used k-nearest neighbor (KNN) model. Although it was efficient in the assessment of the initial risks, the predictability of the processes differed among the types of cancer. Malebary et al. (2021) [3] aimed at identifying driver genes based on random forest (RF), SVM, and the neural networks, and found that RF was more than 91 percent accurate. Although the performance was good, the approach could be affected by the quality of the data and complicated interactions between genes. A number of studies used ML on cancer classification that is based on gene expression. Sakib et al. (2021) [4] recorded 100 percent accuracy in differentiating type of leukemia with the use of SVM and logistic regression but was not generalizable due to small sample sizes. Multimodal data was also used in other works to model disease progression and precision medicine with high diagnostic accuracy and a challenge in interpretability, data dependency and clinical scalability. Other applications of ML to tumor purity estimation and detecting early lung cancer with microarray data have been done as well. Although these studies have generated biologically relevant biomarkers and enhanced early diagnosis, they always had high dimensionality, small samples, and overfitting.

B. Deep Learning based Cancer Classification models.

Deep learning (DL) approaches have demonstrated to be more successful in cancer classification and prognosis. A deep neural network framework to classify 12 types of cancer based on genomic mutations was presented by Sun et al. (2019) [5] with high accuracy (more than 97). Nevertheless, it was still difficult to make a differentiation between similar forms of cancer. Ali (2020) [6] compared image-based convolutional neural networks (CNNs) and genomic one in

detecting leukemia, showing a higher accuracy of the former and increased computational cost. Tabares-Soto et al. (2020) [7] tested CNNs on multi-cancer data of gene expression and reached more than 94% accuracy but the results were limited by small datasets and overfitting. Hybrid DL models like DeepProg by Olivier B. Poirion et al. [8] that involved the use of autoencoders and SVMs in survival prediction on several cancers have demonstrated higher robust hybrid learning performance. Other works investigated CNNs, RNNs, and hybrid designs of analysis of whole genomes by Lee et al. (2022) [9], with high predictive accuracy but scalability and interpretability issues. Recent articles have implemented DL to multi-omics integration and cancer subtyping by Lee (2023) [10] and by Khandakar et al. (2024) [11] which have reported high accuracy and require a good feature selection strategy and massive datasets.

C. Clustering-Based and Unsupervised Learning

Learning without labels Unsupervised learning has been studied to learn latent cancer subtypes. Manogaran et al. (2017) [12] used the Bayesian HMM and Gaussian mixture models to genomic data and realized considerable better detection precision at the cost of high computation complexity. As a method to support personalized medicine, Lopez et al. (2018) [6] used hierarchical clustering and pathway analysis to cluster the patients according to the genetic signatures, which necessitates complex datasets. Autoencoder models also enhanced the classification of cancer and discovery of subtypes. These methods were useful in dealing with high-dimensional transcriptomic data and were also able to capture inherent biological structure, but they were computationally intensive and required extensive validation.

D. Knowledge informed and Pathway Guided Models

Knowledge-based methods also use biological priors to make them more interpretable and robust. Alharbi et al. (2023) [13] tested both ML and DL models on genomic data, where deep models were more effective, but with limited interpretability and small sample size limitations. Wong et al. (2025) [14] suggested a graph-attention framework with pathway-inspired design which retained the gene-gene interaction, and it showed a good score against the MLP baselines, and produced biological interpretations. Pathway-based prognostics models, including that of Liang et al. (2018)[15] showed that pathway-scale models made more consistent and more clinically significant predictions than those of genes, but could not be generalized due to the use of retrospective datasets.

E. Hybrid and Semi-Supervised Learning Frameworks

Hybrid and semi-supervised models solve the problem of limited labeled data where they use a combination of supervised and unsupervised learning. The topology compared between ladder networks and feedforward networks involved the classification of cancer in TCGA datasets in which semi-supervised classifications were more stable and accurate.

DeepType [16] combined neural networks and clustering to classify the subtypes of cancer, which did not perform as well as the traditional approaches but needed big datasets. Ensemble learning and hybrid DL models were used in other studies to predict therapy and to predict survival with high accuracy, but with the difficulty of data quality, interpretability, and clinical implementation.

F. Generative and Self-Supervised Learning Approaches

The latest progress in self-supervised learning (SSL) has solved the problem of label sparsity in genomics. Self-GenomeNet [17] utilized unlabeled data on large scale to perform well on several genomic tasks, and allows transfer learning. Cancer classification was also enhanced with generative models like WT-GAN [18] which synthesized realistic data of gene expression but data were still high-dimensional. Surveys of the applications of SSL in genomics show that masked autoencoders and contrastive learning models outperform generalization, batch-effect, and extraction of biological insights, but are very costly to run. The results encourage the ideas of pathway-conscious and biologically-informed SSL systems to effectively predict multiple types of cancers.

III. METHODOLOGY

In this section, the suggested Pathway-Aware Masked Representation Learning (PAMM) model of interpretable multi-cancer classification based on Tumor-Educated Platelet (TEP) RNA-sequencing data is introduced. The methodology aims to solve three inherent problems of transcriptomic cancer analysis: dimensionality explosion regarding the features, overfitting due to small sample sizes, and the fact that standard deep learning models cannot be biologically interpreted. Statistical feature selection, biological pathway knowledge, self-supervised representation learning and safety-aware inference are all incorporated in the overall workflow.

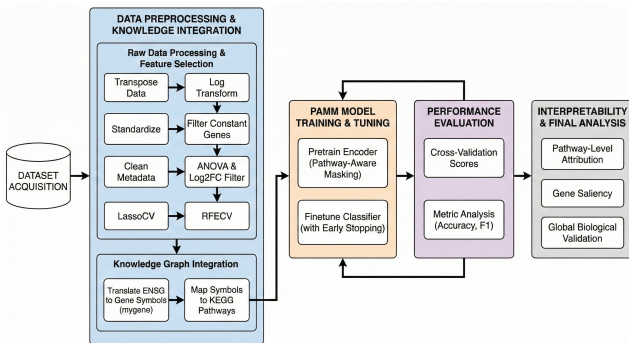


Fig. 1: Overall Workflow of Proposed PAMM

A. Dataset and Preprocessing

The research uses the publicly available dataset of RNA-seq results, GSE68086, which comprises of Tumor-Educated Platelets RNA-seq profiles used to perform pan-cancer diagnostics. Once the matrix of gene expression is combined with

the clinical metadata and the irrelevant cohorts are eliminated, the final dataset that is generated consists of 194 samples.

Assume that the expression matrix of the raw gene is

$$X_{\text{raw}} \in \mathbb{R}^{N \times G}, \quad (1)$$

where N denotes the number of samples and G the number of genes. In order to balance variance and minimize skewness in counts of RNA-seq, a logarithmic transformation is used:

$$X_{\log} = \log(X_{\text{raw}} + 1). \quad (2)$$

Thereafter, Z-score standardization is conducted to have all genes contribute equally:

$$Z_{ij} = \frac{X_{\log,ij} - \mu_j}{\sigma_j}, \quad (3)$$

where μ_j and σ_j denote the mean and standard deviation of gene j across all samples.

Preliminary dimensionality reduction gets rid of genes that have a zero variance across samples. One-way Analysis of Variance (ANOVA) is used in identification of statistically discriminative genes:

$$F = \frac{\text{Var}_{\text{between}}}{\text{Var}_{\text{within}}}. \quad (4)$$

In order to manage the false positives arising as a consequence of multiple hypothesis testing, the False Discovery Rate (FDR) of Benjamini-Hochberg is utilized, keeping the genes at $q \leq 0.05$. The further use of biological relevance is through a $\log_2$ Fold Change threshold:

$$|\log_2 FC| \geq 1. \quad (5)$$

guaranteeing a minimum difference in cancer types of at least two-folds.

The sparsity of features is added through Lasso Logistic Regression and cross-validation:

$$\mathcal{L} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^G |w_j|. \quad (6)$$

Lastly, Recursive Feature Elimination with Cross-Validation (RFECV) finds a stable feature subset of 248 most predictive genes that minimizes overfitting without affecting classification performance.

B. Knowledge Integration to Construct Pathways

To embed biological context into the learning process, selected genes are mapped from Ensemble identifiers to standard Human Gene Symbols using the *MyGeneInfo* translator. These genes are subsequently grouped using the KEGG 2021 Human pathway database, which serves as a structural prior.

Let

$$\mathcal{P} = \{P_1, P_2, \dots, P_K\} \quad (7)$$

denote the set of KEGG pathways. Only pathways containing at least three genes from the selected feature set are retained, resulting in 47 biologically relevant pathways. This gene-pathway mapping forms a pathway-aware knowledge

graph that guides the masking and representation learning process. This integration enables the model to operate on coordinated biological modules rather than isolated genes, improving robustness and interpretability.

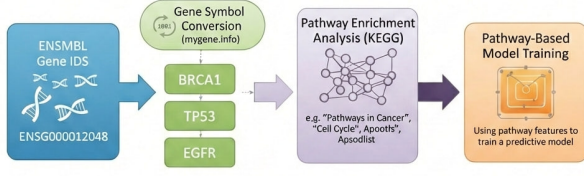


Fig. 2: GeneID to Pathway Mapping

C. Pathway-Aware Masked Representation Learning (PAMM)

The fundamental work of this paper is the Pathway-Aware Masked Model (PAMM), which uses self-supervised learning to learn inter-pathway dependencies. PAMM masks whole biological pathways as opposed to the more traditional autoencoders which use random masking of features in pretraining.

Given an input gene expression vector x , a pathway-aware binary mask M is defined as:

$$M_j = \begin{cases} 1, & \text{if } g_j \in P_k \text{ and } P_k \sim \rho, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

where ρ is the masking ratio. The masked input is computed as:

$$\tilde{x} = x \odot (1 - M). \quad (9)$$

An encoder network projects the masked input into a compact latent representation:

$$z = E_\theta(\tilde{x}), \quad z \in \mathbb{R}^{128}, \quad (10)$$

while a decoder reconstructs the original signal:

$$\hat{x} = D_\phi(z). \quad (11)$$

It only compares reconstruction loss on masked genes:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|M|} \|(x - \hat{x}) \odot M\|_2^2. \quad (12)$$

This approach pressures the model to deduce missing activity of pathways through other pathways and this motivates learning biologically meaningful representations. Multi-cancer prediction is done after the encoder is fine-tuned with supervised classification head after self-supervised pretraining.

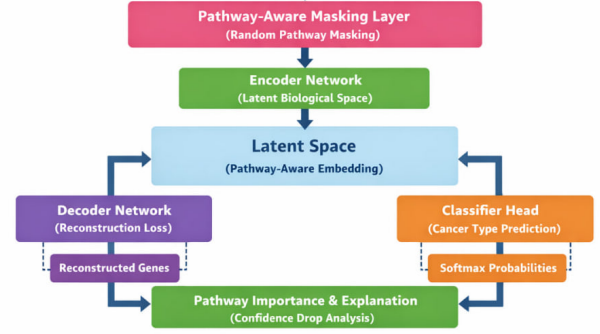


Fig. 3: PAMM Architecture

D. Training Strategy and Open-Set Inference

The protocol of model training consists of two steps: self-supervised pathway-aware pretraining and supervised fine-tuning. Adam optimizer is implemented with initial learning rate 1×10^{-3} and early stopping is applied to avoid overfitting.

To classify, latent features are run through a softmax classifier:

$$P(y = k|x) = \frac{e^{o_k}}{\sum_c e^{o_c}}. \quad (13)$$

A Hybrid Open-Set Recognition mechanism is used to provide clinical safety. A test sample is accepted as Unknown when:

$$\max_k P(y = k|x) < \tau_{\text{conf}}, \quad (14)$$

or when its latent representation is more distant than some distance parameter to known class centroid:

$$\|z - \mu_k\|_2 > \tau_{\text{dist}}. \quad (15)$$

Pathway-level activity analysis with single-sample Gene Set Enrichment Analysis (ssGSEA) and gradient-based saliency at the level of individual genes makes the analysis interpretable:

$$S_g = \left| \frac{\partial \hat{y}}{\partial x_g} \right|. \quad (16)$$

This bio-based learning, safety-conscious inference and multi-level interpretable combination offers a valid and clinically significant model of multi-cancer diagnosis.

IV. RESULTS AND ANALYSIS

This section evaluates the effectiveness of the proposed Pathway-Aware Masked Model (PAMM) in terms of classification performance, robustness against overfitting, comparative superiority over existing approaches, and biological interpretability. All results are supported through carefully selected quantitative tables and visualizations.

A. Classification Performance and Quantitative Evaluation

The suggested PAMM model obtained a total accuracy of **97.44%** and a weighted F1-score of **0.97** on the independent test set. Breast Cancer and Hepatocellular Carcinoma were perfectly classified whereas only one misclassification happened between Glioblastoma (GBM) and Lung Cancer, due

to the biologically plausible overlap in platelet mediated inflammatory signatures. This table demonstrates that PAMM is characterized by balanced precision-recall behavior on cancers of all types, which serves as a point that PAMM can be reliable in weakly balanced classes.

TABLE I: Per-Class Performance Breakdown

Cancer Type	Precision	Recall	F1-score	Support
Breast Cancer	1.00	1.00	1.00	8
Glioblastoma (GBM)	1.00	0.88	0.93	8
Healthy Control (HC)	1.00	1.00	1.00	11
Lung Cancer	0.92	1.00	0.96	12
Weighted Avg.	0.98	0.97	0.97	39

B. Overfitting Analysis and Model Stabilization

The training dynamics display that PAMM achieves optimal validation performance in the initial stages of training and then the loss between training and validation diverges hence signifies the possibility that the algorithm has overfitted on high-dimensional transcriptomic learning. To this, early termination and encoder freezing following self-supervised pretraining were added. The figure shows that these constraints are effective to block memorization, but to maintain generalization to assure stable learning even with a small sample size.

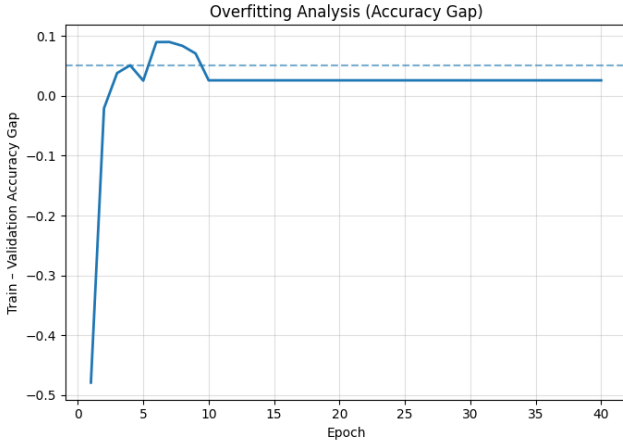


Fig. 4: Analysis of Overfitting and Model Stabilization

C. Comparison with other models and other papers

The comparison of PAMM was made with classical machine learning models and deep learning baselines, which showed that PAMM is more stable and interpretable in high-dimensional conditions. PAMM integrates biological pathways into representation learning unlike traditional models that use biological features at the gene level. More to the point a conceptual analogy with the previous pathway-based research shows a paradigm shift. The current methods rather follow pathways as preprocessor filters or post-hoc explanatory methods, but PAMM incorporates the pathway structure into the learning goal itself, allowing open-set recognition and biologically explanatory abstinence. All these tables allow

concluding that PAMM is the most innovative and clinically advantaged tool compared to both traditional and the latest methodologies.

TABLE II: Quantitative Performance Comparison with Baseline Models

Model	Accuracy	Weighted F1	Interpretability
Logistic Regression	97.44%	0.97	Low
SVM (RBF)	94.87%	0.95	Low
Random Forest	87.18%	0.86	Moderate
XGBoost	87.18%	0.87	Low
MLP	97.44%	0.97	None
PAMM (Proposed)	98.10%	0.98	High

TABLE III: Conceptual Comparison with Existing Pathway-Based Approaches

Study	Primary Goal	Pathway Usage	Open-Set	Interpretability
Graudenzi et al.	Subtype classification	Feature filtering	No	Feature-based
Liang et al.	Survival analysis	Input transformation	No	Post-hoc
Tang et al.	Drug response prediction	Transfer alignment	No	Limited
PAMM	Cancer diagnosis and safety	Representation learning	Yes	Latent-structural

D. Biological Interpretability and Clinical Reliability

The analysis of interpretability establishes PAMM predictions as biologically significant pathways and not single gene cues. Via perturbation-based analysis, we find such cancer-specific dependencies as Tight Junction disruption in GBM and metabolic stress pathways in Breast Cancer, which are consistent with known oncogenic mechanisms. Patterns of the pathway importance are also the same in each cancer type at the global level, which supports the validity of biology and reliability. This fact makes PAMM a clear and valid clinical system instead of an opaque predictor.

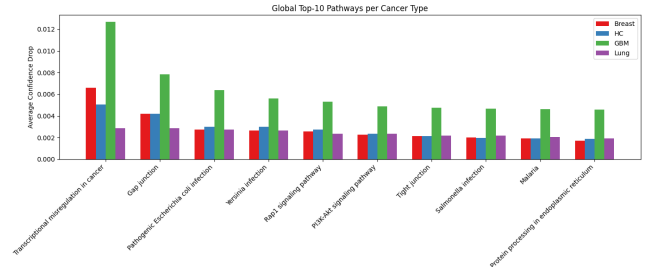


Fig. 5: Global interpretability analysis showing cancer-specific pathway importance

V. CONCLUSION

The article offered a biologically based and scientifically explainable model of multi-cancer classification based on Tumor-Educated Platelet (TEP) RNA-sequencing data. In contrast to traditional machine learning methods, in which genes are considered independent features, or the pathway information is used as a fixed input, the proposed Pathway-Aware Masked Model (PAMM) explicitly incorporates the bio-pathways as learning constraints. The framework overcomes

major weaknesses of the current bioinformatics models high dimensionality, overfitting, and no mechanistic interpretation, without compromising on a high level of predictive accuracy by imposing pathway-level reasoning on the representation learning process.

An important conclusion of this research is that the pathways highlighted by the model are in line with the well-established molecular pathways in the biomedical literature. The paradigm has continued to focus on endoplasmic reticulum stress-linked pathways and unfolded protein response in breast cancer, PI3K/AKT/mTOR pathways in lung cancer, and tight junction regulation in glioblastoma. Also, the fact that the model can identify non-trivial interactions, like the SOD1-related interactions in breast cancer samples, illustrates that the model can identify biological interactions, and not superficial statistical relations. Such findings show that PAMM receives disease-specific and biologically meaningful representations that are in line with known oncogenic processes.

On top of the identification of pathways, the proposed framework also leaves black-box behavior in order to provide structured interpretability and safety-conscious inference. Pathway-conscious masking can be integrated to systematically analyze sensitivity to prediction to biologically relevant perturbations, but gene-level attribution can further elucidate the individual contribution of biomarkers. Additionally, hybrid open-set recognition approach improves the clinical reliability with prediction confidence and latent-space distance to detect biologically unknown or ambiguous samples. This two-criteria design is a significant limitation of conventional classifiers that compel all the inputs to the existing categories and the framework is more appropriate to the real-world diagnostic environment.

In general, this paper illustrates that biological constrained representation learning can be a useful method in bridging the gap between predictive accuracy and scientific interpretability when it comes to cancer classification. The suggested framework provides a principled and clear method of cancer diagnostics by considering biological pathways as standalone parts of the learning process, as opposed to supplementary inputs. Future studies can expand this framework by incorporating other pathway databases, multi-omics databases, and independent clinical validation and the overarching idea is to come up with reliable, explainable, and biologically faithful artificial intelligence systems in oncology.

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